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This PDF contains only a brief excerpt of the work, as I am currently awaiting arXiv preprint endorsement to upload the complete version. I hope this snippet is helpful.

Thank you.

Knowledge-Based Systems

Plaque Burden and Carotid Intima-Media Thickness Measurements in Ultrasound Scans: Are Transformer Tuners a Must for UNet Architectures?

--Manuscript Draft--

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Corresponding Author:	jasjit suri, Ph.D. Atheropoint, Roseville California UNITED STATES
First Author:	Nikhil Singh
Order of Authors:	Nikhil Singh Ayman El-Baz Arun Kumar Dubey Laura Eve Mantella Amer M Johri Esma R Isenovic Sanjay Saxena Narendra N Khanna Mustafa Al Maini Vijay Viswanathan Gavino Faa Andrew Laine Luca Saba jasjit suri, Ph.D.
Abstract:	Abstract Background and Motivation: UNet-based Deep Learning systems have dominated in the field of medical imaging for segmentation of organs; however, it is not accurate, validated, or generalized. Transformers or attention, on the other hand, provide solutions that are sophisticated but use heavy-duty computations and take a long time to converge. This study introduces a two-stage segmentation solution that combines transformers as tuners to the base UNet, resulting in a highly accurate and generalizable solution. Method: Three sets of base UNets were designed, namely, B1:UNet, B2: UNet++, and B3:UNet+++, and four sets of tuners were designed, namely: T1:Transformer-augmented UNet, T2:Attention-guided UNet, T3:Swin Transformer-based UNet, and T4:Pyramid-based network, leading to 12 fusion systems that combine three base UNets and four Tuners, namely: B1+T1, B1+T2, B1+T3, B1+T4; B2+T1, B2+T2, B2+T3, B2+T4; B3+T1, B3+T2, B3+T3, B3+T4. A comparison is made on 13 important metrics, including model size, complexity, training time, inference speed, and accuracy-to-parameter efficiency. Results show that B1+T4, which achieved the highest score of 100%, was the best-performing system. Results: Findings indicate that the two-stage segmentation model is effective and reliable than the single-stage UNet architecture. The combination of single UNet backbones and transformers/attention mechanisms then creates a structure-sensitive and extremely nuanced pipeline. Altogether, the proposed hybrid framework is scalable, high-throughput, and fits properly into the real-life clinical practice where the accuracy and stability of segmentation vital importance. Conclusions: We conclude that Hybrid transformer-based models like B1+T4 provide highly accurate, reliable, and automated techniques that segments and measure the

risk of CVD.

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Nikhil Singh^{1,2}, Ayman El-Baz³, Arun K. Dubey¹, Laura E. Mantella⁴, Amer M. Johri⁵,
Esma R Isenovic⁶, Sanjay Saxena⁷, Narendra N. Khanna⁸, Mustafa Al-maini⁹, Vijay Viswanathan¹⁰,
Gavino Faa^{11,12}, Andrew Laine¹³, Luca Saba¹⁴, Jasjit S. Suri^{2,15,16,17}

¹Bharati Vidyapeeth's College of Engineering, New Delhi, INDIA.

²Stroke Monitoring and Diagnostic Division, AtheroPoint™, Roseville, California, USA.

³Department of Bioengineering, University of Louisville, Kentucky, USA.

⁴Division of Cardiology, Department of Medicine, Queen's University, Kingston, ON, CANADA.

⁵Department of Biomedical and Molecular Sciences, Queen's University, Kingston, ON, CANADA.

⁶Department of Radiobiology and Molecular Genetics, "VINČA" Institute of Nuclear Sciences - National Institute of the Republic of Serbia, University of Belgrade, Belgrade, SERBIA.

⁷Department of CSE, IIIT, Bhubaneswar, INDIA.

⁸Department of Cardiology, Indraprastha APOLLO Hospitals, New Delhi 110001, INDIA.

⁹Allergy, Clinical Immunology and Rheumatology Institute, Toronto, ON M5G 1N8, CANADA

¹⁰MV Diabetes Centre, Royapuram, Chennai, Tamil Nadu, INDIA.

¹¹Department of Medical Sciences and Public Health, University of Cagliari, A.O.U. of Cagliari, Cagliari, ITALY.

¹²Department of Biology, College of Science and Technology, Temple University, Philadelphia, PA, USA.

¹³Departments of Biomedical and Radiology, Columbia University, New York, NY, USA.

¹⁴Department of Radiology, Azienda Ospedaliero Universitaria (A.O.U.) di Cagliari, Polo di Monserrato, Cagliari, ITALY.

¹⁵Department of ECE, Idaho State University, Pocatello, ID, 83209, USA.

¹⁶University Center for Research & Development, Chandigarh University, Mohali, INDIA.

¹⁷Symbiosis Institute of Technology, Nagpur Campus, Symbiosis International (Deemed University), Pune, INDIA.

*Corresponding Author:

Dr. Jasjit S. Suri, PhD, MBA, FIEEE^a, FAIMBE^b, FSVM^c, FAIUM^d, FAPVS^e

^aFellow, The Institute of Electrical and Electronics Engineers

^bFellow, American Institute of Medical and Biological Engineering

^cFellow, Society of Vascular Medicine

^dFellow, American Institute of Ultrasound in Medicine

^eFellow, Asia Pacific Vascular Society

Stroke Monitoring and Diagnosis Division

AtheroPoint™, Roseville,

Roseville, CA 95661, USA

Phone: (916)-749-5628

Email: jasjit.suri@atheropoint.com

Author Contributions

Conceptualization, N.S. and J.S.S.; Methodology, N.S.; Software, N.S.; Validation, A.E.B., L.E.M., A.M.J., S.S., N.N.K., M.A.-M., V.V. and J.S.S.; Investigation, G.F., A.L., L.S., and J.S.S.; Resources, J.S.S.; Data curation, A.K.D.; Writing—original draft, N.S.; Writing—review & editing, N.S., E.R.I, and J.S.S.; Visualization, N.S., and J.S.S.; Supervision, A.K.D, and J.S.S. All authors have read and agreed to the published version of the manuscript.